

Danial Al-Ghazali Walangadi

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Research Statement

I am interested in leveraging state-of-the-art AI technologies to address computer security challenges, including issues related to privacy, security, accountability, and the sociotechnical aspects of cybersecurity.

Education

Universitas Negeri Semarang, BSc in Computer Science 2023 – 2027 (expected)

- Overall GPA: 3.92/4.0 (Transcript)
- Major GPA: 3.92/4.0
- Coursework:

Operating Systems (A), Algorithms and Programming (A), Introduction to Computer Science (A), Computer Architecture and Organization (AB), Computational Logic (A), Data Structures (A), Algorithm Analysis (A), Human-Computer Interaction (A), Computer Networks (A), Discrete Mathematics (A), Database Systems (A), Formal Languages and Automata Theory (A), Artificial Intelligence (A), Numerical Methods (A), Object-Oriented Programming (A), Image Processing (A), Software Engineering (A), Information Technology Research (AB), Compiler Design (AB).

Research Experience

System and AI Research Training Program (SYAIR) Aug 2025 – Dec 2025

- Indonesia's international research training program instructed by **Prof. Haryadi Gunawi** from the University of Chicago.
- Selected as one of the top 50 student researchers in the Computer Science field nationally.
- Reviewed more than 10 scholarly papers presented at leading computer science conferences, including but not limited to ACM CCS, NDSS, and EuroS&P.
- Analyzed codebases that are available from the supplementary materials of reproducible conference papers.
- Successfully replicated and thoroughly understood the full implementation of a conference paper on backdoor purification for malware classifiers using a fine-tuning method in the field of computer security.
 - Assessed and reproduced critical experiments of "**PBP: Post-training Backdoor Purification for Malware Classifiers**" NDSS 2025. (Reproduction Report[↗])

Projects

Strengthened AES-128 Encryptor Github[↗]

- Scaffolded a single-page web application that provides an implementation of AES-128 encryption alongside a reproduction of a research article that aims to strengthen the AES system by using novel S-Boxes.
- Designed the underlying AES-128 logic with TypeScript's standard library without any external libraries.
- Provided the logic for generating AES keys, the encryptor-decryptor mechanism, dynamic AES S-Box usage, and relevant testing environments using Vitest.
- Overall source code length is over **2,500 LOCs** without counting Vite's React initialization overhead.
- The implementation was created with React, deployed with Cloudflare Pages, and is publicly available for use under the Apache 2.0 License.
- Tools Used: **TypeScript, React, TailwindCSS, Vitest, Cloudflare Pages**

Toy 16-bit RSA Encryptor Github[↗]

- Built a simple web interface that provides an example of how RSA encryption works.
- Designed the underlying RSA logic with TypeScript's standard library without any external libraries.

- Constructed the logic for generating RSA-relevant variables, RSA keys, and the encryptor-decryptor mechanism.
- Overall source code length is over **230 LOCs** without counting Vite's React initialization overhead.
- The implementation was created with React, deployed with Cloudflare Pages, and is publicly available under the Apache 2.0 License.
- Tools Used: **TypeScript, React, Cloudflare Pages**

License Plate Extractor

Kaggle[↗]

- Created a deep learning model, allowing users to extract the text from images containing a license plate.
- The project was trained under Kaggle's T2 GPUs, with the codebase spanning over **740 LOCs**.
- The source code is available publicly under the 2.0 version of the Apache License.
- Tools Used: **PyTorch, YOLO, Pandas, Matplotlib**

Ngiprit Money

Github[↗]

- Developed a web interface to keep track of transactions, debts, receivables, and their relevant statistics.
- The project comprises over **3,000 LOCs** without counting the overhead introduced by Laravel.
- The application can be used freely under version 3.0 of the GNU Affero General Public License.
- Tools Used: **Laravel, React, TailwindCSS, ChartJS, Date-fns, PostgreSQL**

Technical Skills

Languages: C++ , Python, TypeScript, LaTeX

Operating Systems: Windows, Debian WSL, Ubuntu

Cloud Platform: Digital Ocean, Chameleon Cloud

Embedded Systems Hardware: ESP32, Orange Pi

Machine Learning Frameworks: PyTorch

Web Development: React, TailwindCSS, Laravel

Desktop Applications: Tauri

Database Systems: PostgreSQL, SQLite

Language Proficiency

Indonesian Language

June 2022

Native speaker

UKBI: 387/800 (Madya)

English

June 2022

EFSET: 82/100 (CEFR C2)

Certification & Training

- Cisco Networking Academy Certified Computer Networking Skills (2024)
- ANC Japan Japanese Language Course (2025)

References

Prof. Haryadi Gunawi

Computer Science, University of Chicago

Research Program Instructor

Abas Setiawan, S.Kom., M.Cs.

Computer Science, Universitas Negeri Semarang

Lecturer, Academic Advisor